

Fresh Fare Distribution Co. - Analytics Case Study

Company Background

Fresh Fare Distribution Co. (FFDC) is a mid-sized food distribution company operating in five major states across the United States (CA, NY, TX, FL, IL). Founded in 2015, FFDC specializes in supplying fresh produce, dairy, meat, frozen goods, and beverages to restaurants, cafes, hotels, and catering businesses. The company prides itself on maintaining optimal temperature control throughout its supply chain and ensuring timely deliveries to maintain food quality.

Business Context

In 2023, FFDC experienced rapid growth but also faced challenges in maintaining service quality and optimizing operations. The company operates through:

- 5 temperature-controlled warehouses
- 10 delivery drivers
- Fleet of 5 refrigerated delivery vehicles
- Serving approximately 100 regular customers
- Managing inventory of 50 different products across various categories

Current Situation

The management team wants to analyze their 2023 operational data to:

1. Improve delivery performance
2. Optimize inventory management
3. Understand customer behavior
4. Maximize profitability
5. Plan for strategic expansion

Available Dataset

The company maintains six key transactional tables:

1. **Orders**
 - Basic order information

- Customer details
- Order dates and expected delivery dates
- Payment terms
- Total order values
- 2. **OrderDetails**
 - Line-item level information
 - Product quantities
 - Unit prices
 - Discounts applied
 - Line-item totals
- 3. **Products**
 - Product specifications
 - Storage requirements
 - Temperature thresholds
 - Pricing information
 - Supplier details
- 4. **Customers**
 - Business information
 - Contact details
 - Location data
 - Credit limits
 - Customer type
- 5. **Deliveries**
 - Delivery tracking
 - Driver assignments
 - Temperature logs
 - Actual delivery times
 - Delivery status
- 6. **InventoryTransactions**
 - Stock movements
 - Batch tracking
 - Expiry dates
 - Warehouse assignments
 - Transaction types

Business Questions for Analysis

Sales and Revenue Analysis

1. Which customer segments contribute most to our revenue, and how has their ordering pattern evolved through 2023?

2. What is the correlation between discount rates and order volumes across different customer types?
3. How do seasonal factors affect sales across different product categories?

Operational Efficiency

1. What is our current on-time delivery rate, and how does it vary by:
 - Geographic region?
 - Customer type?
 - Product category?
2. How effective is our temperature control during deliveries, and what impact does it have on customer satisfaction?
3. Which routes or delivery schedules demonstrate the best efficiency?

Inventory Management

1. What is our current inventory turnover rate by product category?
2. How often do we experience stockouts or near-stockout situations?
3. What is the optimal reorder point for each product considering delivery times and demand patterns?

Customer Insights

1. Can we identify at-risk customers based on ordering patterns and delivery performance?
2. What characteristics define our most profitable customers?
3. How does credit limit utilization correlate with order frequency and size?

Financial Performance

1. What is our gross profit margin by:
 - Product category?
 - Customer segment?
 - Geographic region?
2. How do transportation costs impact profitability across different delivery routes?
3. What is the financial impact of our current discount strategy?

Required Analysis

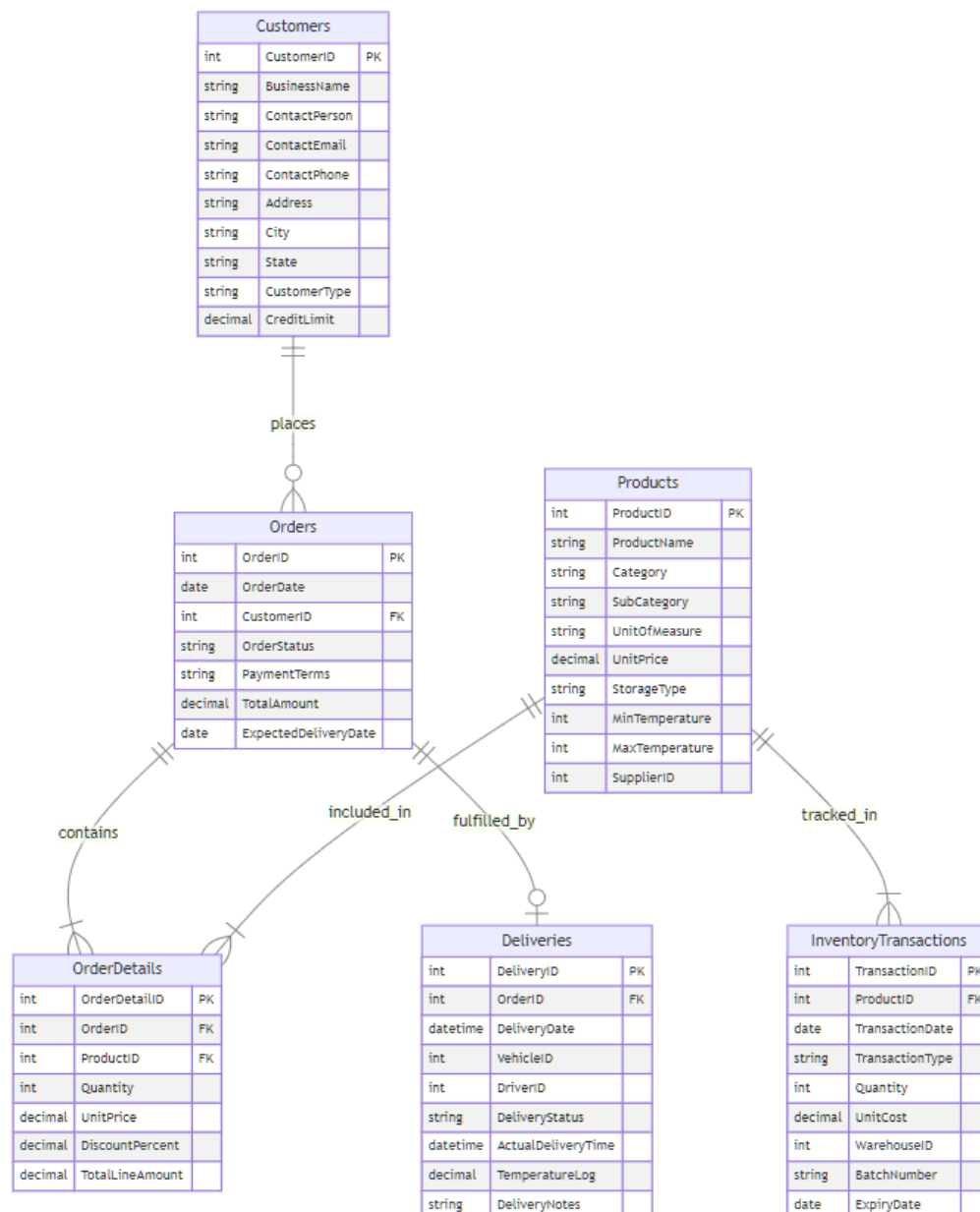
As a data analyst, you are required to:

1. Design appropriate metrics and KPIs

2. Create a dimensional model for efficient analysis
3. Develop analytical queries to answer business questions
4. Provide actionable insights and recommendations
5. Identify areas for operational improvement

Data Details

- Time Period: January 2023 - December 2023
- Transaction Volume: ~1000 orders
- Number of Customers: 100
- Product Range: 50 SKUs
- Geographic Coverage: 5 states



Tables with columns

-- Orders placed by customers (restaurants/cafes)

Orders

- OrderID (PK)
- OrderDate
- CustomerID
- OrderStatus
- PaymentTerms
- TotalAmount
- ExpectedDeliveryDate

-- Individual line items in each order

OrderDetails

- OrderDetailID (PK)
- OrderID (FK)
- ProductID
- Quantity
- UnitPrice
- DiscountPercent
- TotalLineAmount

-- Product inventory transactions

InventoryTransactions

- TransactionID (PK)
- ProductID
- TransactionDate
- TransactionType (Inbound/Outbound)
- Quantity
- UnitCost

- WarehouseID
- BatchNumber
- ExpiryDate

-- Delivery tracking

Deliveries

- DeliveryID (PK)
- OrderID (FK)
- DeliveryDate
- VehicleID
- DriverID
- DeliveryStatus
- ActualDeliveryTime
- TemperatureLog
- DeliveryNotes

-- Customer master

Customers

- CustomerID (PK)
- BusinessName
- ContactPerson
- ContactEmail
- ContactPhone
- Address
- City
- State
- CustomerType
- CreditLimit

-- Product master

Products

- ProductID (PK)
- ProductName
- Category
- SubCategory
- UnitOfMeasure
- UnitPrice
- StorageType
- MinTemperature
- MaxTemperature
- SupplierID